

Reframing Money: Blockchain-Native Tokens as Endogenous Drivers of Innovation and Growth in Decentralized Economies

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Abstract: Studies on cryptocurrencies have predominantly focused on their role as financial infrastructure and their impact on existing monetary systems, while overlooking the institutional logic that positions cryptocurrencies as a new form of money. Many analyses still adhere to the classical “monetary veil” paradigm, which underestimates cryptocurrencies due to their perceived instability in fulfilling the traditional functions of money—unit of account, medium of exchange, and store of value. This paper challenges this view by proposing that decentralized, permissionless blockchain-native tokens align with Joseph Schumpeter’s Credit Theory of Money (CTM), particularly in their role in fostering entrepreneurial innovation. We introduce the concept of the Productive Deflationary Token Regime, in which seigniorage is directed toward ecosystem expanders and incentives are stabilized through a deflationary design. By reshaping issuance and reward structures, this regime allows blockchain ecosystems to maintain a continuous, non-extreme flow of token supply and to remain operational across boom–bust cycles, rather than letting incentives expand and collapse with short-term price movements. This approach enables the active organization of production and market growth, independent of state-backed or centralized banking systems. Unlike traditional credit money, the Productive Deflationary Token Regime supports decentralized, endogenous economic expansion and innovation, representing a blockchain-driven evolution of Schumpeter’s model of innovation mobilization in the digital economy.

Keywords: cryptocurrency, blockchain, Schumpeter, decentralized economy, endogenous economic growth

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I. Introduction

The emergence of cryptocurrency is often regarded as a response to the 2008 global financial crisis. Bitcoin, the first cryptocurrency, was introduced in 2009 by the pseudonymous Satoshi Nakamoto, whose core proposition was to establish a peer-to-peer electronic cash system based on blockchain technology, eliminating reliance on centralized institutions such as governments and banks by removing the need for intermediary trust (Nakamoto, 2008). Moreover, Bitcoin employs cryptographic algorithms to ensure transaction anonymity and asset security, thereby technologically reinforcing the protection of individual rights. Overall, Nakamoto's vision was to create a monetary system composed solely of equal individuals in a free market, without the involvement of states or banks.

However, whether cryptocurrencies can truly function as decentralized, non-sovereign money remains a subject of significant academic debate. On the one hand, lacking state-backed credit and not being anchored to any physical asset, cryptocurrencies struggle to maintain public confidence in their monetary status (Chey, 2023). Their high price volatility has led to their more frequent classification as speculative financial assets (Yukun Liu, 2021; Hu and Zhang, 2023), with many scholars questioning their ability to perform the three traditional functions of money and to compete with fiat currency systems (Ametrano, 2016; IMF, 2018: 52; Brunnermeier, 2019; Leo Malherbe, 2019; Marco Fama et al., 2019; Arıkan, 2020; Au, 2023). On the other hand, there is a broad consensus that technological innovation alone is insufficient to establish the legitimacy of money; the creation and operation of money must be embedded within political, social, and legal institutional structures (Hutten and Thiemann, 2018). In fact, rather than diminishing the dominance of states and centralized banks within the monetary system, blockchain technology has arguably reinforced their position, constituting another round of technological empowerment for existing state monetary systems and centralized financial institutions (Beaumier and Kalomeni, 2022; Rella, 2019; Maurer, 2016; Campbell-Verduyn and Goguen, 2019; Brandl & Dieterich, 2023), thereby limiting the value of privately issued cryptocurrencies.

These critiques reflect the extent to which current understandings of cryptocurrency remain constrained by existing political economy frameworks. First, our conceptualization of money has long been shaped by the “monetary veil” paradigm inherited from classical economics, which examines cryptocurrency through the lens of the classical monetary functions—unit of account, medium of exchange, and store of value (Frasser & Guzman, 2020, p.1). Within this framework, the potential institutional innovations of cryptocurrency are often overlooked. Second, academic discussions have primarily focused on the political and economic implications of blockchain technology as a novel financial infrastructure within the existing sovereign monetary and financial system, while paying comparatively little attention to cryptocurrency itself as an independent monetary form (Chey, 2023).

In recent years, as fiat-pegged stablecoins have gained an increasing share of the cryptocurrency market—and, owing to their practical advantages in cross-border settlement and payment efficiency, have attracted widespread attention from central

banks, regulatory agencies, and academia—political economists have begun to move beyond the classical theory of money. They have increasingly turned to an alternative framework, the Credit Theories of Money (CTM)³, to analyze cryptocurrencies and their monetary systems as a novel form of money. Within the CTM framework, “moneyness” is rooted in a social relationship—trust—where money is understood as a special form of credit. The central question, however, is how such trust is initially established and how a monetary instrument comes to be widely accepted (Minsky, 2008, p. 228)⁴. At present, the prevailing academic view draws an analogy between stablecoins and unregulated forms of traditional credit money, conceptualizing stablecoins as a type of “shadow money” whose functional and risk characteristics resemble those of money market mutual funds (Allen, 2022; Anadu et al., 2023; FSB, 2022; Li & Mayer, 2022; Aldasoro et al., 2023). Yet beyond stablecoins, the operational mechanisms of truly “decentralized” cryptocurrencies as a new monetary form remain insufficiently theorized.

A recent study by Olk and Miebs (2025) applies the CTM framework to cryptocurrencies other than stablecoins—such as Bitcoin, Ethereum, and certain DeFi platform tokens—and argues that they constitute a form of “new shadow money.” Although these cryptocurrencies originally emerged as “anti-credit money,” their acquisition of monetary functions has depended on the rise of “credit intermediaries” such as centralized exchanges (CEXs) and stablecoins. In essence, such institutions function as shadow banks, thereby introducing new credit mechanisms into the system. As a result, rather than achieving the goal of “de-creditization,” cryptocurrencies have become embedded within a shadow banking structure.

In contrast to both the objects of analysis and the conclusions of Olk and Miebs (2025), this paper focuses on blockchain-native tokens—including Bitcoin, Ethereum, and Solana—and argues that they embody an institutional logic that requires no trust, yet functionally parallels the “credit expansion—innovation mobilization” mechanism emphasized in Joseph Schumpeter’s CTM. In Schumpeter’s framework, banks initiate new production processes and drive innovation and economic growth by extending loans and granting credit to entrepreneurs; the essence of money lies in its capital function—its capacity to mobilize resources for new production and development—and the process of money creation is an institutional arrangement endogenous to innovation and economic growth (Schumpeter, 1954; Messori, 1997). This paper contends that blockchain networks nurture a similar monetary regime, one that does not

³ This classification stems from two major cleavages in the history of monetary thought: the first is the divide between Monetary Theories of Credit (MTCs) and Credit Theories of Money (CTMs). The former defines money before deriving credit, and whether in state-based MTCs (money as a legal construct) or market-based MTCs (money as a commodity), it emphasizes that money must simultaneously fulfill its three functions. The latter, by contrast, places credit before money, viewing money as a specific form of credit and focusing more on its unit-of-account function (Murau & Pforr, 2023). Joseph Schumpeter, in his *History of Economic Analysis* (1954), was the first to systematically elaborate on the opposition between MTCs and CTMs.

⁴ As Hyman Minsky put it, ‘everyone can create money’, but ‘the problem is to get it accepted’ (Minsky, 2008, p. 228).

rely on bank-created credit but instead sustains token value and drives market expansion through endogenous incentive mechanisms.

The central research question of this paper is: What is the source of value for blockchain-native tokens that are neither backed by state sovereignty, nor pegged to physical assets, nor reliant on centralized bank-created credit? Furthermore, as a novel monetary form, what are their functions within blockchain networks, and through what mechanisms do they operate? This paper conceptualizes such tokens as constituting a Productive Deflationary Token Regime. These tokens are not merely “rewards” for productive activities within the blockchain network—such as mining, liquidity provision, and protocol development—but also act as mechanisms of mobilization. At the core of this regime is the distribution of seigniorage from productive deflationary tokens as rewards to validators, liquidity providers, developers, and other contributors to the blockchain ecosystem. The effectiveness of these rewards, however, depends on whether the token’s value can be sustained or even increased over the long term. Crucially, token value is not derived from state-backed credit or a physical value peg, but from the ongoing expansion of the ecosystem and the sustained growth in demand for block space. Thus, to ensure the long-term effectiveness of such production incentives, productive tokens must be capable of mobilizing market expansion. The deflationary mechanism is the key to sustaining this mobilization effect. Rather than simply creating scarcity to drive short-term price appreciation, it operates by adjusting the pace of token supply and the intensity of rewards—strengthening incentives during bear markets, while moderately reducing rewards during bull markets to prevent producers from over-extracting ecosystem value. This design constitutes a form of “countercyclical” incentive structure, ensuring that, in a freely competitive token market, the network can continuously attract producers, maintain essential productive capacity, and expand the ecosystem. In sum, blockchain networks with native productive deflationary tokens represent a new type of market order, in which token-based incentives guide market expansion. The productive deflationary token thereby functions as a monetary institution capable of sustaining economic order and endogenously driving economic growth—without relying on state sovereignty or bank-created credit.

The main contributions of this paper are as follows. First, it introduces the Productive Deflationary Token Model for the first time, demonstrating that in a freely competitive token market, native tokens have the potential to endogenously drive innovation and market expansion, thereby challenging conventional monetary theories’ understanding of the economic functions of cryptocurrencies. Second, it constructs a systematic dataset by collecting and organizing information from blockchain project websites, white papers, and on-chain transaction data, and by using large language models to automatically extract textual parameters related to token supply mechanisms. These parameters are then standardized and structured to provide a data foundation for subsequent empirical research.

The remainder of the paper is organized as follows: Section 3 reviews relevant monetary theories; Section 4 presents and elaborates the theoretical framework of the Productive Deflationary Token Model; Section 5, based on the constructed dataset,

traces the evolution of token supply mechanisms and changes in market structure; Section 6 conducts empirical tests of the model's core mechanisms using supplementary on-chain data; and Section 7 concludes with a summary and discussion.

II. Review of Monetary Theories

In the study of International Political Economy (IPE), three major monetary theories are most relevant to analyzing the nature of cryptocurrencies: the commodity theory of money (also known as the metallist theory), the state theory of money (STM), and the credit theory of money (CTM). The first two generally interpret money within the framework of its three classical functions—medium of exchange, unit of account, and store of value—differing primarily in the relative priority assigned to each function. The credit theory of money, by contrast, breaks from this logic and does not require that money fulfill all three functions simultaneously. This paper argues that both the commodity theory and the state theory have limitations in explaining the value origins and functions of cryptocurrencies. In comparison, the Schumpeterian perspective within the credit theory of money offers a more suitable analytical framework for understanding the functions and operational logic of this new monetary form within blockchain networks.

1. Commodity Theory of Money

The commodity theory of money can be traced back to Aristotle (Aristotle, 1995) and was inherited by classical economists such as Adam Smith (Smith, 1976 [1776]). This theory posits that money originated from market-exchangeable commodities (such as metals or shells) that came to serve as media of exchange, thereby overcoming the high transaction costs of barter (Bell, 2001: 151–152; Ingham, 2004: 57). Within this framework, Bitcoin is often described as “digital gold” due to its predictable and scarce supply ensured by algorithmic design, which is seen as granting it anti-inflationary properties and the potential to function as a store of value. However, it has also been criticized for imposing macroeconomic constraints similar to those of the historical gold standard in the international financial system. More importantly, the commodity theory's explanatory framework often overlooks the institutional functions of cryptocurrencies.

2. State Theory of Money

The state theory of money, developed by Knapp (1905), defines money as a “chartal means of payment” created by the state (Knapp, 2003 [1924]: 38). Its essence lies in establishing a uniform unit of account and requiring citizens to use that unit for taxation and public payments (Ingham, 2004: 33, 57). Although blockchain networks are not state entities, their native tokens are likewise designated—by network protocol—as the unit of account and medium of payment within the network, reflecting a form of “digital sovereignty” whereby a decentralized network establishes its own monetary regime. These tokens are primarily used to pay network gas fees, thus functioning as the unit of account for blockchain block space (as will be elaborated later). Nevertheless, due to their high price volatility, their capacity to serve as a stable medium of exchange in the real economy remains contested.

3. Credit Theory of Money

The credit theory of money (CTM) encompasses Keynesianism, the Schumpeterian school, institutionalist economists, as well as more recent frameworks such as the *Money View* and *Modern Monetary Theory*. Although there is internal disagreement within CTM over whether money is ultimately determined by the state or by the market (Murau & Pforr, 2023), proponents generally understand money as a credit relationship embedded in specific payment contexts, rather than as the direct product of a particular commodity or a state decree (Innes, 1913; Minsky, 1986). Once a credit-debt relationship arises within a payment system, it may, under certain conditions, serve as a medium of exchange for some participants in that system⁵. Consequently, defining money solely as an instrument that functions as a medium of exchange in *all* contexts is a misunderstanding of the operational logic of the modern monetary system (Murau & Pforr, 2023). Within this framework, the “moneyness” of cryptocurrency does not depend on whether it simultaneously fulfills all three classical functions in every setting, but rather on whether it can generate a form of trust within a given payment system such that it is accepted as the unit of account and medium of settlement in that context.

Joseph Schumpeter’s *Credit Theory of Money* goes beyond traditional conceptions of money, emphasizing that the essence of money lies not in its role as a store of value or a convenient medium of exchange, but in its capital function—its capacity to initiate new production processes and drive economic development. In this framework, banks create money through credit expansion, enabling entrepreneurs to mobilize resources in advance for innovative investment. The banking system thus ceases to be a passive financial intermediary and becomes an active creator of “monetary capital,” with lending itself constituting the generation of purchasing power, thereby fueling dynamic economic growth (Schumpeter, 1912; 1934; 1939). Schumpeter’s theory is therefore a quintessential dynamic theory of money: money is no longer a neutral medium of exchange in a static equilibrium, but an institutional arrangement endogenous to the processes of innovation and economic growth (Schumpeter, 1954; Messori, 1997).

III. A Growth-Oriented Issuance Framework: Rewarding Market-Expanding Agents

1. Costs and Sources of Value of Productive Tokens

A productive token refers to a token whose ongoing issuance is driven by productive activities—where the actions of certain participants generate monetary supply and seigniorage is distributed to them as a reward for their productive contributions. A typical example is the block reward generated in the process of block production. In such models, newly issued tokens are generally distributed to block producers—miners or validator nodes—according to the consensus mechanism in place.

⁵ However, not all credit-debt instruments must serve as a medium of exchange for all potential counterparties. In different contexts and at different points in time, the acceptability of particular instruments as a means of discharging debts varies. Within a hierarchy of money, instruments that function as the *final means of settlement* and occupy higher positions in the hierarchy possess a greater degree of “moneyness.”

Under the Proof of Work (PoW) mechanism, nodes (i.e., miners) must expend substantial computational resources to solve cryptographic hash puzzles in competition for block production rights. The first node to compute a valid hash value receives the reward for that block. This process reflects not only the significant resource investment and the cost of token production but also ensures a direct correspondence between token rewards and productive activity. Under the Proof of Stake (PoS) mechanism, the acquisition of block production rights is linked to the quantity of tokens staked by a node and its protocol-compliant behavior. Unlike the physical energy consumption in PoW, the primary resource cost in PoS lies in the *opportunity cost* of staked tokens. That is, in order to obtain block production rights, a node must lock up a certain amount of tokens, which cannot be used for other purposes—such as investment, liquidity provision, or arbitrage trading—during the staking period. The potential forgone returns thus constitute the production cost of tokens under PoS. More broadly, the issuance of productive tokens can be tied to a variety of productive activities beyond block production, including protocol development and liquidity provision. In these cases, participants receive token rewards in exchange for contributing critical resources, and the resources committed constitute the cost of token production.

Accordingly, when the issuance of productive tokens entails significant production costs, producers will only engage in such production if their *expected returns* exceed those costs. For miners, such returns typically consist of three components: newly issued token rewards, MEV (maximal extractable value), and transaction fees⁶. At present, the first two account for the majority of expected returns. The magnitude of token rewards is determined directly by the number of tokens distributed and their market value. The value of blockchain tokens derives from users' demand for block space, which can only be accessed by holding and spending tokens. Each token issued corresponds to the creation of a certain share of block space. The generation of block space, in turn, requires tangible resource inputs—whether computational power, the opportunity cost of staked tokens, or liquidity. This means that productive tokens are anchored to actual productive activities and resource commitments. In other words, productive tokens represent the value of producers' contributions, granting them the right to claim corresponding compensation from the network based on their production efforts, while simultaneously providing a measurable, transferable unit that can serve as a final means of payment for block space.

In contrast, a non-productive token refers to a token whose supply mechanism, over its entire lifecycle, is never directly linked to any productive activity. Such tokens are often fully pre-minted at the time of chain launch and subsequently released in stages through vesting schedules, commonly in the form of linear vesting or cliff vesting. These mechanisms are widely applied to the initial allocations for project teams,

⁶ Block rewards: These represent a form of seigniorage, or a forward-looking claim on the future value of the token. Maximal Extractable Value (MEV): MEV offers additional revenue by allowing validators to strategically reorder, include, or censor transactions within a block. This form of income is particularly significant in programmable environments like Ethereum, where DeFi arbitrage, liquidations, and front-running opportunities are prevalent (Daian et al., 2020).

early investors, or foundations. Although vesting introduces a time-based release period, the distribution logic depends solely on the passage of time rather than on recipients' sustained contributions to the network, thus lacking any institutional linkage to ecosystem activity. Issuance mechanisms for non-productive tokens also include airdrops, typically employed as a user acquisition or community incentive tool, with distribution logic more akin to a marketing campaign. Other models involve genesis-based fixed supply, as in the case of Stellar, where the total token supply is minted at launch and no further issuance or unlocking occurs thereafter. Another variant is discretionary issuance, in which future supply decisions are delegated to a foundation or core developers. While this approach offers considerable flexibility, its issuance logic remains disconnected from productive activities. Overall, tokens that are issued at zero cost and not tied to productive behavior often lack a solid basis for value. Without sustained price support from market makers, such tokens are prone to “pump-and-dump” dynamics—experiencing a one-off price surge followed by rapid collapse.

2. Survival Logic of Productive Tokens: Ecosystem Expansion

The continued survival of a productive token depends on whether its underlying blockchain network can sustain genuine and ongoing market demand within a freely competitive environment. The high degree of openness among blockchain ecosystems—combined with the low technical costs for users and developers to migrate—means that a blockchain network hosting a productive token must maintain the capacity for ecosystem expansion in order to preserve sustained demand for its block space. Once a network loses vitality, faces stagnation in its use cases, or suffers from a lack of applications, users are likely to migrate rapidly to alternative networks that are more efficient or better suited to their needs. Such ecosystem-level liquidity not only influences token demand—and thus token value—but also affects the scope for miner/maximal extractable value (MEV), a key source of returns for producers. When market demand declines, token value shrinks, undermining the production incentive mechanism and triggering a chain reaction of reduced network security and ecosystem contraction. Accordingly, the essence of the productive token regime is not simply to provide a static “reward” for individual productive behavior, but to serve as a mechanism for organizing production and mobilizing market expansion. Newly issued tokens (i.e., seigniorage) are distributed to key actors who participate in ecosystem expansion and bear early-stage risks. This mechanism tightly couples token issuance to ecosystem growth, making market creation an intrinsic driver of token issuance. Thus, productive tokens not only fulfill functions of value measurement and payment but also embody an economic organization function. Through incentives, they coordinate individual actions and guide capital, developers, and users to converge on the network—operating as a mobilization mechanism oriented toward market expansion.

In practice, large validator nodes often invest in infrastructure, wallet development, and ecosystem tooling to encourage adoption and stimulate token demand. For instance, Chorus One offers institutional-grade staking services on networks like Cosmos, Ethereum, and Solana, while also developing cross-chain tools such as Staking SDK

and OPUS Pool to lower the entry barrier for users⁷. Figment, a leading cross-chain validator, provides APIs (e.g., DataHub) and staking indexers to enable unified access to multiple blockchains, thereby facilitating decentralized application (DApp) development and ecosystem expansion.⁸

3. Purpose of the Deflationary Mechanism: Sustaining Incentive Stability and Continuous Ecosystem Expansion

The capacity of productive tokens to incentivize production and mobilize market expansion relies on a deflationary mechanism. The core objective of this mechanism is to maintain the relative stability of the incentive structure rather than the stability of token value itself (indeed, in some cases it may be designed to increase token value). In some productive token models, the deflationary mechanism takes the form of gradually reducing block rewards, ensuring that producers' earnings from production are not perpetually diluted by unlimited issuance. More importantly, this design motivates producers to expand the market continuously, thereby increasing token value and enlarging the scope for MEV (maximal extractable value) to offset the decline in block rewards, as in Bitcoin's halving mechanism or Solana's annual reduction in inflation rates.

Other productive tokens employ supply-adjustment mechanisms to stabilize incentives across changing market conditions. Specifically, as shown in Table 1, productive deflationary tokens often constrain rewards (by lowering inflation or increasing burn rates) during market booms to prevent producers from over-extracting system value. Conversely, they relax this constraint (by modestly raising inflation or reducing burn rates) during downturns—for example, in the case of Ethereum's EIP-1559—in order to prevent the collapse of the incentive mechanism and the interruption of productive activity. Through this continuous adjustment of the incentive structure, the system is able to avoid both explosive issuance in euphoric phases and a sudden shutdown of rewards in slumps. In practice, this yields a smoother and more sustainable ecosystem-wide supply path over time, preserving baseline productive capacity and retaining the potential for ecosystem expansion across market cycles. By contrast, productive tokens lacking a deflationary mechanism typically exhibit much stronger volatility in their incentive schedules: their inflation tends to surge in bull markets and collapse in bear markets. In such systems, the pace of issuance is not anchored in ecosystem demand, but is instead largely driven by short-term market sentiment and contemporaneous price movements, which exacerbates reward dilution in booms and leaves producers without institutional support in downturns, thereby increasing the long-term risk of ecosystem contraction.

In essence, the deflationary mechanism functions to calibrate token supply by reshaping incentives so that, over time, the path of token issuance remains broadly compatible with the ecosystem's long-run demand for security, liquidity, and development, rather than fluctuating mechanically with market mood or token price. This compatibility underpins the stability of the productive token's incentive structure:

⁷ <https://everstake.one/events>

⁸ <https://figment-docs.gitbook.io/learn-docs/guides/datahub-products>

an excessively rapid pace of supply growth leads to value leakage from the ecosystem, while an excessively slow pace of issuance undermines network maintenance—both ultimately contributing to ecosystem decline. Unlike traditional monetary regimes, which primarily aim at value stability, productive deflationary token systems embody a fundamentally different logic: they prioritize a sustainable incentive structure and a viable long-run supply path over short-term price stability. This institutional logic can be interpreted as a decentralized evolution of Schumpeter’s credit theory of money. In Schumpeter’s framework, banks extend credit to innovators, enabling structural economic transformation and growth. In blockchain systems, productive deflationary tokens achieve a similar outcome without bank-issued credit: they direct seigniorage toward ecosystem expanders and use the deflationary mechanism to maintain incentive stability and a continuous, non-extreme flow of token supply across market conditions, thereby sustaining ongoing market expansion and constructing an endogenous growth model independent of both state sovereignty and centralized banking systems.

Table1 Inflation Dynamics and Incentive Structures Under Market Cycles

	Bull Market (High Demand)	Bear Market (Weak Demand)
Productive & Deflationary	Highly competitive block space, high network activity, high transaction fees, and high token value → more token burning / dynamic block reward reduction	Decreased transaction demand, low fees, low network activity, lower token value → reduced burning, dynamic reward increase
	Feature: Incentives are driven by high fees, high MEV, and high token value; block rewards are reduced	Feature: Fee and MEV incentives are lacking, token value per unit drops → block rewards are increased to compensate
Productive & Non-deflationary	No deflationary constraint, active market, relatively high token value → high block/staking rewards → inflation rate rises	Weak network, low activity, relatively low token value → low block/staking rewards → inflation falls
Non-Productive	Often distributed via airdrops or unlocked after pre-mining, without relying on productive activities. The issuance schedule is more often linear or governed by preset rules, lacking dynamic adjustment mechanisms. As a result, the inflation rate is merely a function of time, not of market conditions.	

IV. The Evolution of Blockchain Cryptocurrency Supply Mechanisms

Before delving into the empirical analysis, it is necessary to clarify the terminological divergence between the crypto world and the traditional macroeconomic discourse on monetary policy. In conventional macroeconomics, the concepts of *inflation* and *deflation* are closely tied to changes in the price level. Inflation is typically understood as a general increase in the prices of a representative basket of goods and services over time, reflecting rising living costs and a decline in the purchasing power of money (Öner, 2017). Conversely, deflation refers to a persistent decrease in the overall price level (Akerlof, 1996), often associated with weak aggregate demand,

overproduction, monetary tightening, or capital flight. In this framework, inflation and deflation are judged with reference to changes in real-world prices, typically anchored to a fiat currency unit and a concrete set of goods and services.

In contrast, within the domain of crypto tokenomics, these terms are largely symbolic or supply-based constructs. Most crypto assets are not pegged to fiat currencies, nor do they operate in an environment where “cost of living” or “consumer price indexes” are meaningful benchmarks. As a result, *inflation* in the crypto context often refers not to price-level dynamics, but to changes in token supply volume (Kane Falco ter Veer, 2024). Many blockchain projects define *inflation rate* as the annual growth rate of token issuance, setting targets such as “a 5% inflation rate,” which in effect denotes the rate at which new tokens are introduced by developers. Under this framing, inflation is equated with the pace of token issuance, rather than with any observable rise in market prices. Similarly, *deflation* does not imply a decrease in price levels across goods or services, but rather a tightening of the token supply—manifested either through token burns, capped supply, or gradual issuance reduction. Many projects strategically construct “deflationary narratives” to emphasize token scarcity, thereby enhancing the expectation of future price appreciation. This narrative, however, is fundamentally supply-side in nature, rather than reflecting actual market price deflation.

Despite the widespread use of inflation and deflation terminology in crypto discourse, the design, articulation, and implementation of token supply mechanisms across projects remain opaque. Moreover, the stated inflation rates often lack economic meaning in the conventional sense. This motivates the present section, which aims to return to the foundational question of how crypto monetary systems are structured. By analyzing how token supply mechanisms are described and implemented across projects, we map the broader evolution of token issuance architectures in the crypto economy. To this end, this study employs a large language model (DeepSeek-V3) to automatically extract structured textual data from the whitepapers and official websites of 269 currently active blockchain projects. Among these, 20 are Layer-2 networks, which are generally excluded from detailed issuance analysis due to their lack of independent token supply mechanisms—being closely integrated with their underlying Layer-1 chains. Therefore, our analysis focuses primarily on the remaining 249 Layer-1 and sidechain networks.

The extracted information covers key aspects of tokenomics, including token issuance, vesting schedules, deflationary design, and staking mechanisms. The majority of the extracted data was manually verified and analyzed to ensure accuracy. For analytical clarity, this study categorizes the token supply mechanisms of blockchain systems into two overarching dimensions: issuance mechanisms and deflationary mechanisms, which together capture the monetary architecture of blockchain-based economies.

1. Issuance mechanisms

Issuance mechanisms refer to how new tokens are introduced into a blockchain network. Based on a survey of 269 blockchain tokens, this study categorizes token supply models into two broad types: productive and non-productive issuance

mechanisms (Figure 1). Productive issuance mechanisms are characterized by a direct linkage between token issuance and block production. Although their initial allocation may include vesting schedules or airdrops, their subsequent supply is contingent upon actions such as block production, staking, liquidity provision, or other verifiable contributions. In contrast, non-productive tokens are native tokens of Layer 1 or sidechain networks whose entire lifecycle of issuance is decoupled from any form of productive behavior (as illustrated in Table2).

Table2 Classification Matrix: Productive vs. Non-Productive Tokens

		Ongoing issuance tied to productive behavior	
		Yes	No
Initial issuance includes pre-minting or airdrops	Yes	Productive Token	Non-Productive Token
	No	Productive Token	/

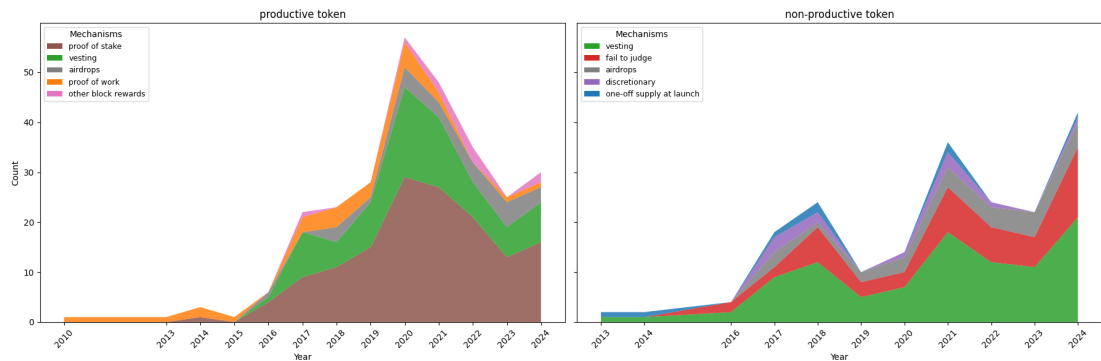


Figure 1. Yearly changes in issuance mechanisms among productive and non-productive token, 2010–2024

Note: The horizontal axis represents the year in which each token was launched, while the vertical axis indicates the number of newly issued tokens that adopted a given issuance mechanism in that year. It is important to note that a single token may incorporate multiple issuance mechanisms; therefore, the total height of the bars may exceed the actual number of tokens launched in a given year.

Figure 1 illustrates the issuance mechanisms of productive and non-productive tokens over time. On the left panel, productive tokens were initially dominated by proof-of-work (PoW) issuance, particularly prior to 2015. Beginning in 2015, proof-of-stake (PoS) became the primary issuance mechanism. From 2017 onward, vesting schedules associated with team, investor, and foundation allocations began to appear among productive tokens. However, these allocations constitute a fixed portion of the total supply; the continued issuance of productive tokens fundamentally depends on ongoing block production, thereby linking token supply to verifiable economic activity within the network. In contrast, the right panel shows that non-productive tokens rely predominantly on vesting-based unlock mechanisms, without being tied to any form of block production. A smaller share is distributed via airdrops, and even fewer tokens are issued through one-off allocations or discretionary issuance. These issuance models are largely exogenous to network usage and do not incorporate endogenous incentives for ecosystem expansion.

2. Deflationary mechanisms

Deflationary mechanisms refer to token supply structures characterized by adjusting token supply either through a reduction in total token supply or a adjustment in the rate of new issuance. These mechanisms can be broadly classified into three categories (as illustrated in Table 3):

Burn Mechanisms: These involve the permanent removal of tokens from circulation, typically through the destruction of transaction fees or periodic token buyback and burn operations. Such practices directly reduce the circulating supply and generate a clear deflationary effect.

Gradual Inflation Reduction Mechanisms: In this model, projects progressively reduce the rate of new token issuance over time. This often takes the form of a declining annual inflation rate or scheduled reductions in block rewards, such as halving events of Bitcoin.

Dynamic Inflation Adjustment Mechanisms: These mechanisms adjust the token issuance rate or reward level in response to on-chain dynamics—such as staking participation and transaction activity—or to underlying network conditions, such as mining difficulty and energy consumption.

Table3 Category of Deflation

Mechanism Name	Definition	Detailed Description in Whitepaper or on the website	Examples
Burn Mechanisms	Mechanisms that aim to control inflation or adjust token supply by permanently reducing the number of tokens in circulation.	Fee burn; Token burn event; Coin burning event; Burned upon production; Buyback and burn; other token burning mechanism; Token Bonding Curve buyback	Eth, Polygon
Gradual Inflation Reduction Mechanisms	Mechanisms that gradually reduce token issuance rates over time to curb inflationary pressure.	Block reward reduction over time; Inflation reduction over time; Deflationary Unlock	Solana, Bitcoin
Dynamic Inflation Adjustment Mechanisms	Mechanisms that adjust inflation or rewards dynamically based on on-chain metrics such as staking rate or network activity.	Current annual reward rate for validators adjusted based on network participation; Annual percentage yield that adjusts based on total staked tokens; Token Bonding Curve	Eth Sei

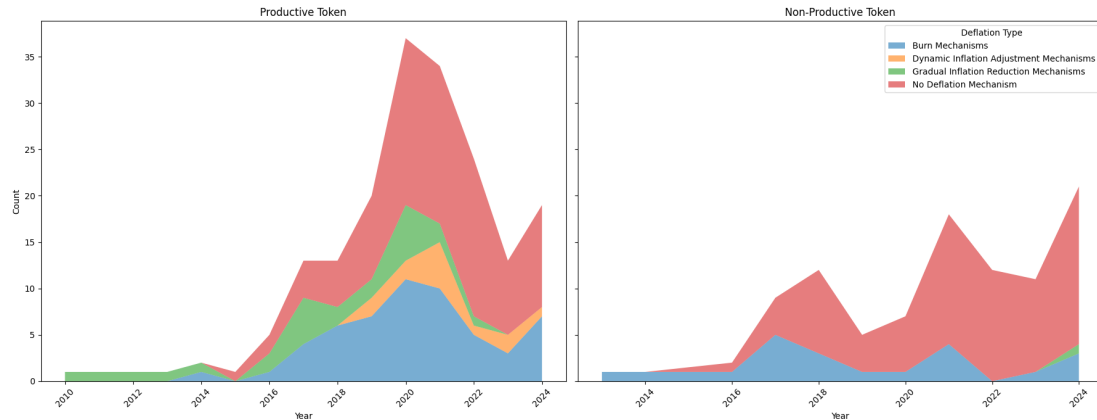


Figure 2. Yearly changes in deflationary mechanisms among productive and non-productive token, 2010–2024

Figure 2 presents the temporal evolution of deflationary mechanisms across productive and non-productive tokens from 2010 to 2024, revealing distinct trajectories in their monetary design. For productive tokens, deflationary mechanisms were present from the early stages of the market. Burn mechanisms and gradual inflation reduction strategies—such as Bitcoin’s programmed block reward halving—were particularly prevalent among early-generation projects. Over time, these designs became increasingly sophisticated: deflationary logic evolved from fixed, rule-based reductions in block rewards to more adaptive, dynamic inflation adjustment mechanisms, which calibrate issuance in response to on-chain metrics, validator participation, or macroeconomic signals. Nonetheless, a notable trend since 2020 is the growing share of newly launched productive tokens that omit any explicit deflationary mechanism, indicating a structural shift in the design logic of token economies. In contrast, non-productive tokens exhibit both a lower prevalence and reduced complexity in the design of deflationary mechanisms. Their deflationary strategies are almost exclusively limited to token burn mechanisms, typically implemented through revenue-funded buybacks or fee-based burning models. Since approximately 2018, the majority of non-productive tokens have omitted any form of deflationary mechanism altogether. This absence is more pronounced than in productive token ecosystems, further underscoring the structural divergence between the two issuance paradigms.

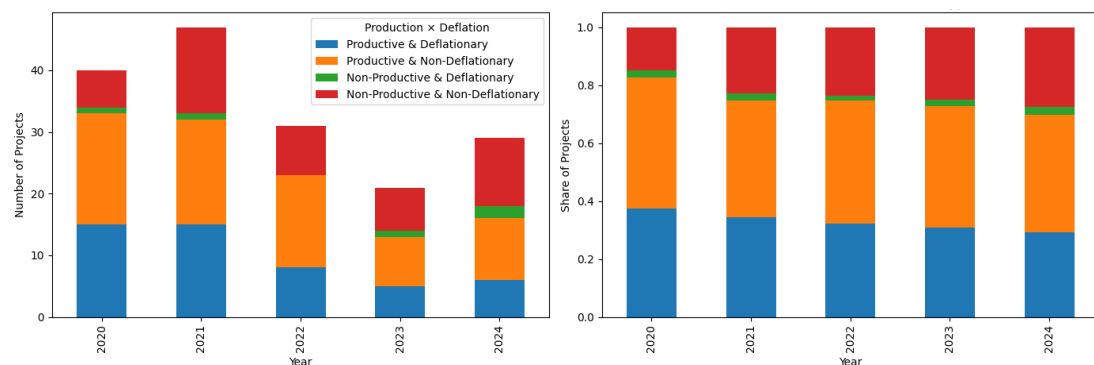


Figure 3. Annual token launches (left) and yearly token supply structure (right), categorized by deflationary mechanism and issuance mechanism, 2010–2024.

Figure 3 shows the annual and cumulative distribution of tokens from 2020 to 2024, categorized by whether their issuance is productive and whether they incorporate a deflationary mechanism. First, the data reveal a steady decline in the market share of productive tokens over time. As shown in the right panel, productive tokens accounted for nearly 80% of the market in 2020, but their share dropped to approximately 60% by 2024, indicating a structural shift toward non-productive issuance models. Second, deflationary design is strongly associated with productive issuance. Tokens that are both productive and deflationary consistently represent 25–30% of the total market, while deflationary non-productive tokens remain below 10% throughout the observed period. This suggests that deflationary mechanisms are disproportionately concentrated in tokens with production-linked issuance, likely because they align better with ecosystem incentives and support long-term value preservation.

V. Empirical Assessment of Growth-Oriented Token Issuance

This paper draws extensively on on-chain data and token trading metrics from platforms such as *defillama*⁹, *stakingrewards*¹⁰, *coinmarketcap*¹¹. The dataset includes time-series observations of token price, market capitalization, trading volume, circulating supply, as well as blockchain-level indicators such as total value locked (TVL), staking yields (for PoS chains), and protocol revenue. These data serve as the empirical foundation for testing the theoretical framework outlined in Section II—namely, the hypothesis that growth-oriented issuance mechanisms, by rewarding market-expanding agents such as validators and miners, contribute to stronger and more sustainable ecosystem development.

1. Token and Ecosystem Performance under Productive and Deflationary Designs

Token Performance. Figures 5, 6, and 7 present token value, standard deviation, and volume distributions from 2020 to 2024 across different issuance design types. Productive and deflationary tokens consistently demonstrate higher median token values and greater trading volumes, indicating stronger market resilience and liquidity. In contrast, non-productive deflationary tokens are limited in number and tend to underperform, suggesting that deflationary mechanisms alone are less effective in the absence of production-linked issuance.

However, productive tokens exhibit greater volatility in token value, showing no clear advantage in terms of price stability compared to other categories. This suggests that the primary function of productive tokens is not to serve as stable media of exchange.

⁹ <https://defillama.com/>

¹⁰ <https://www.stakingrewards.com/>

¹¹ <https://coinmarketcap.com/>

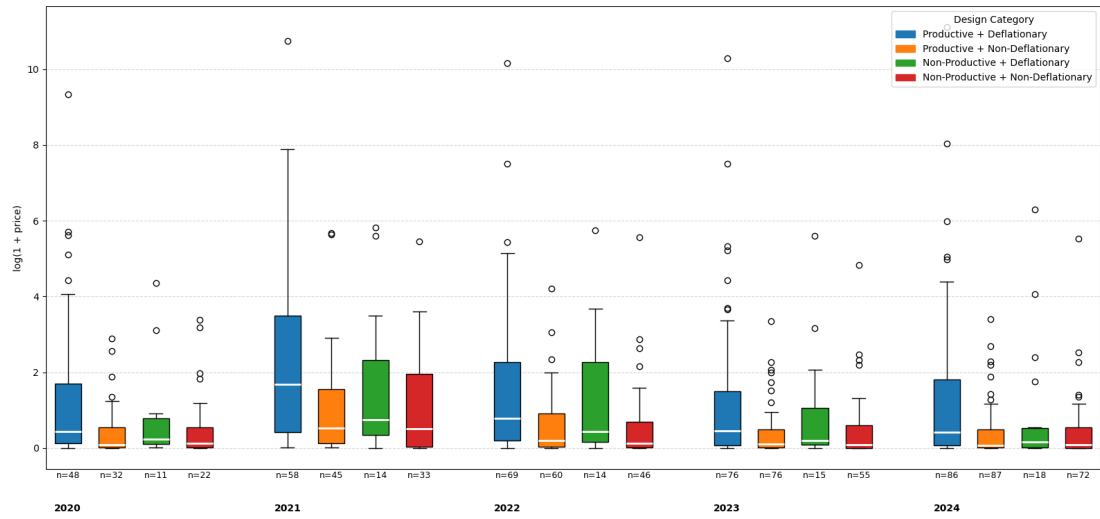


Figure 4. Token Value Distribution, 2020-2024.

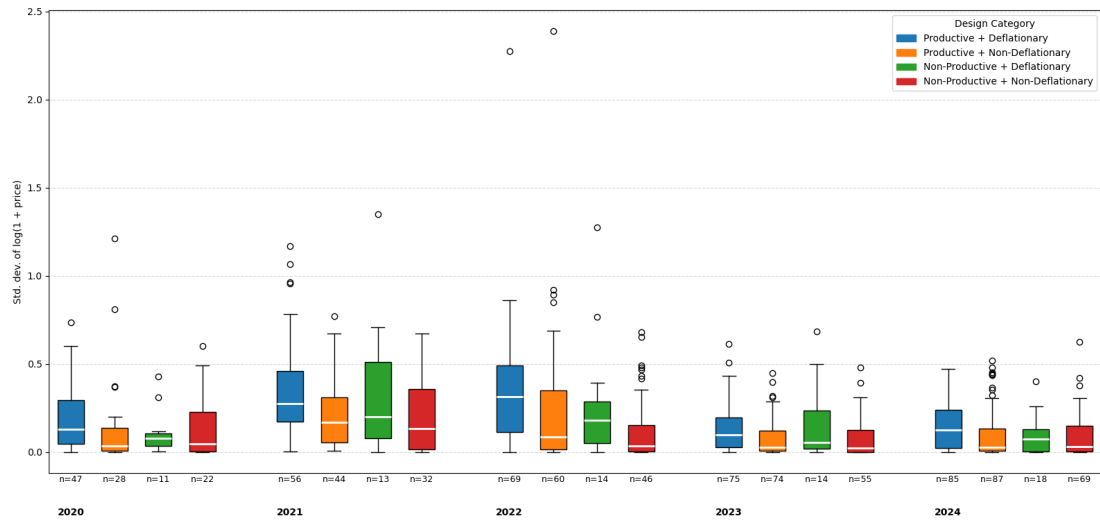


Figure 5. Standard Deviation of Token Value, 2020-2024.

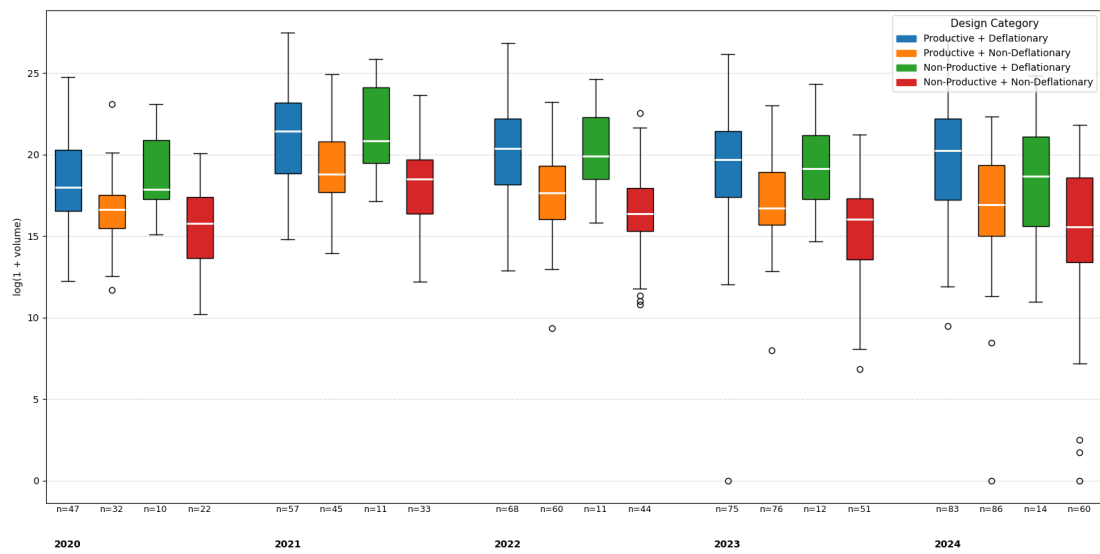


Figure 6. Token Volume Distribution, 2020-2024.

Ecosystem Performance. This paper adopts Total Value Locked (TVL) as the core metric to evaluate the performance of blockchain ecosystems. TVL reflects the actual utilization and immobilization of assets on-chain, serving as a key indicator of a network’s attractiveness, liquidity, and overall ecosystem health. A higher TVL typically implies greater user participation and capital inflow, suggesting that the network is trusted, actively used, and more robust in terms of ecosystem development.

Figure 7 illustrates the changing composition of total TVL across blockchain networks categorized by token design types. Excluding Ethereum and its Layer 2 networks, productive and deflationary tokens have occupied a consistently growing share of total TVL since 2021, demonstrating strong aggregation capacity and value capture potential. In contrast, non-productive deflationary tokens, although maintaining a certain share, show a declining trend over time—highlighting that deflationary mechanisms, when decoupled from productive issuance, may yield limited long-term effects. Tokens without deflationary mechanisms, regardless of whether they are production-linked, remain marginal in terms of TVL contribution.

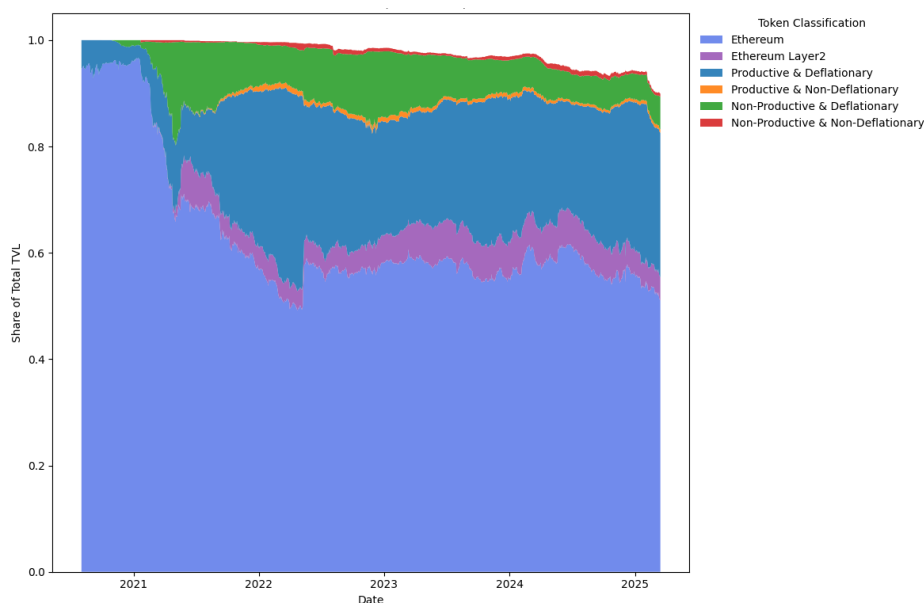


Figure 7. TVL Distribution, 2020-2024.

Note: As in Figure 7, Ethereum and its Layer 2 networks¹² are analyzed separately due to their disproportionately large share of total value locked (TVL) relative to other blockchain networks. It is worth noting, however, that Ethereum is categorized as deflationary, following its transition to Proof of Stake and the adoption of token-burning mechanisms such as EIP-1559.

Given the possibility that Figure 7 may be influenced by extreme values from individual chains, Figure 8 further disaggregates TVL distribution by quartiles to reflect the stratification of ecosystems under different token issuance regimes. The results show that productive-deflationary tokens are significantly concentrated in the top 25%

¹² In my samples, Layer 2 of Ethereum includes Arbitrum, Optimism, Mantle, Starknet, Taiko, ZkFair,

Kroma, LightLink, Ancient8, Zero Network, and Movement.

of ecosystems, with this pattern strengthening in 2023–2024. Figure 9 provides a more granular look at the deflationary mechanisms adopted by the top 20 blockchain networks by TVL. With the exception of BSC, all other leading chains employ productive token models and embed some form of deflationary mechanism—ranging from token burns to dynamic or gradual issuance adjustments. This stratified structure supports the hypothesis that when token issuance is directly linked to productive behavior and governed by a deflationary mechanism, it facilitates more efficient resource mobilization, sustains producer incentives, and promotes positive feedback loops within the ecosystem.

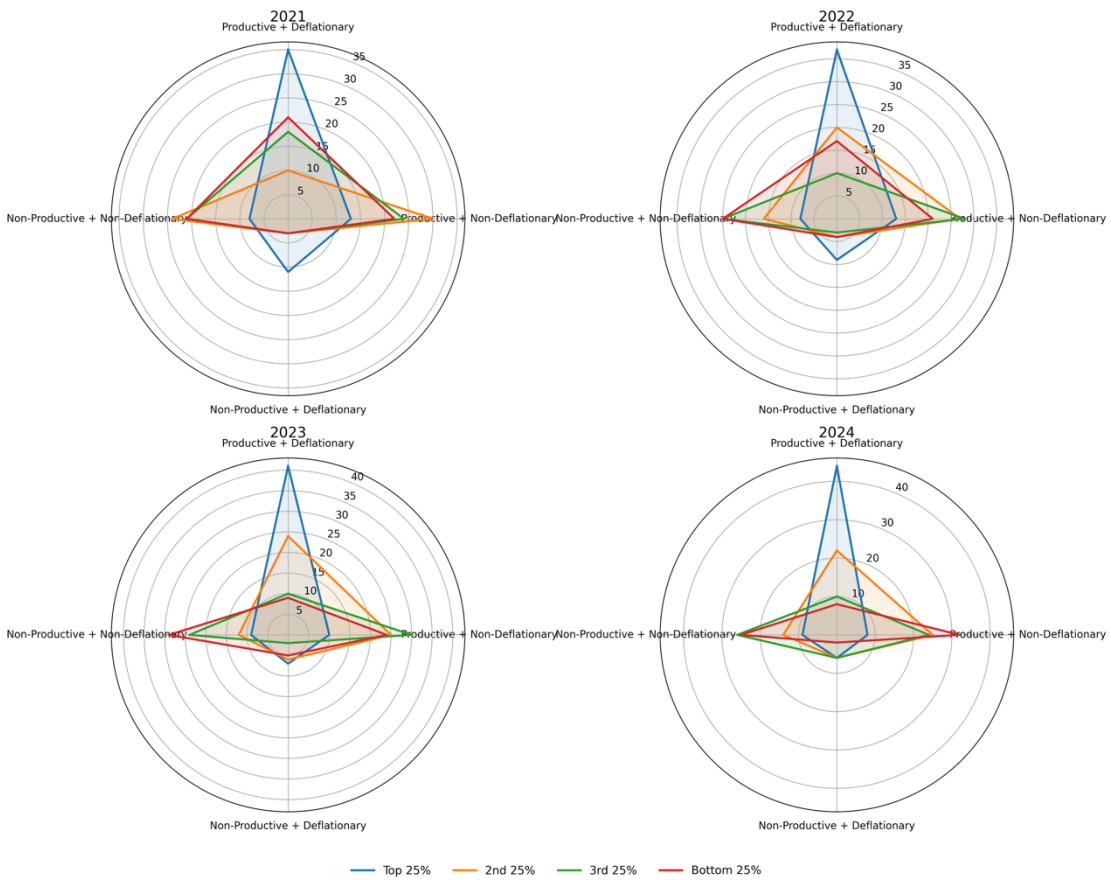


Figure 8. Supply Mechanism Distribution Across TVL Quartiles, 2021–2024.

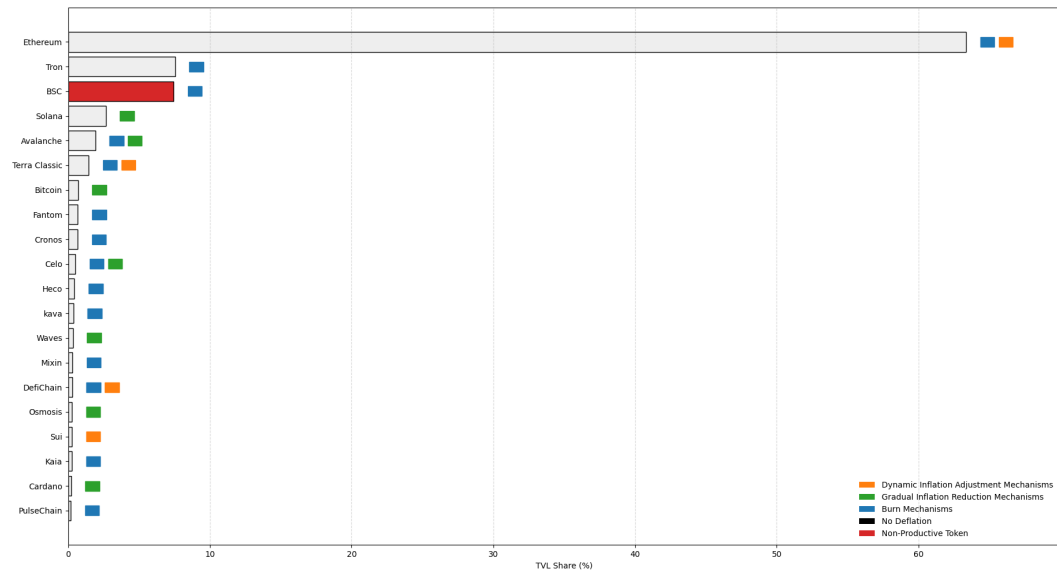


Figure 9. Top 20 Blockchain Networks by TVL and Their Deflationary Mechanisms

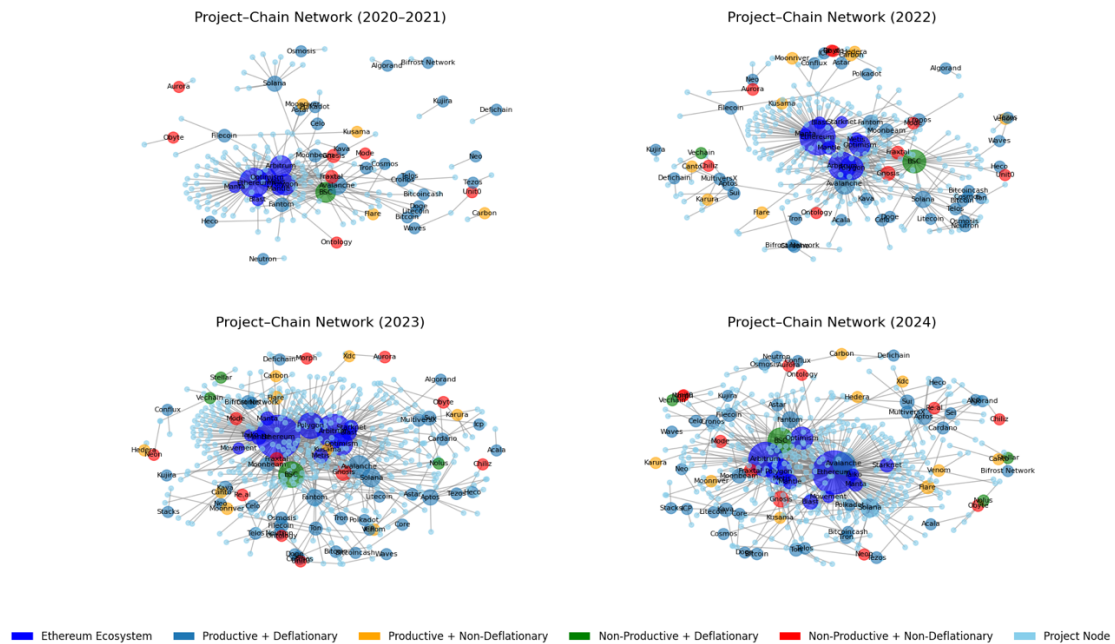


Figure 10. Inter-chain Deployment Network of Decentralized Applications across Blockchain Platforms.

Note: Each small blue dot represents a decentralized application (DApp), which may be deployed on multiple blockchain platforms simultaneously. Larger colored nodes represent blockchains, with node size indicating the number of associated DApps. Node color encodes the native token's supply mechanism. Ethereum and its Layer 2 ecosystem are shown separately in dark blue.

Figure 10 further illustrates the current inter-chain deployment network of decentralized applications across blockchain platforms. First, the Ethereum ecosystem—including its Layer 2 extensions—exhibits a high degree of centrality, with the largest number of protocol deployments. This reflects its strong network effects and sustained developer traction. Second, other blockchains adopting productive and deflationary token mechanisms (e.g., *Solana*, *Avalanche*) also occupy relatively central

positions in the network, with a substantial number of protocol deployments. In contrast, while BNB Chain holds a prominent position in the network, its native token follows a non-productive but deflationary issuance model, suggesting that its ecosystem growth may be more attributable to centralized resource mobilization than to endogenous incentive mechanisms. Other chains characterized by non-productive or non-deflationary token models tend to appear on the periphery of the network, with limited protocol activity, indicating relatively low levels of ecosystem engagement and adoption.

2. Deflation as a Monetary Supply Adjustment Mechanism for Ecosystem Expansion Demand

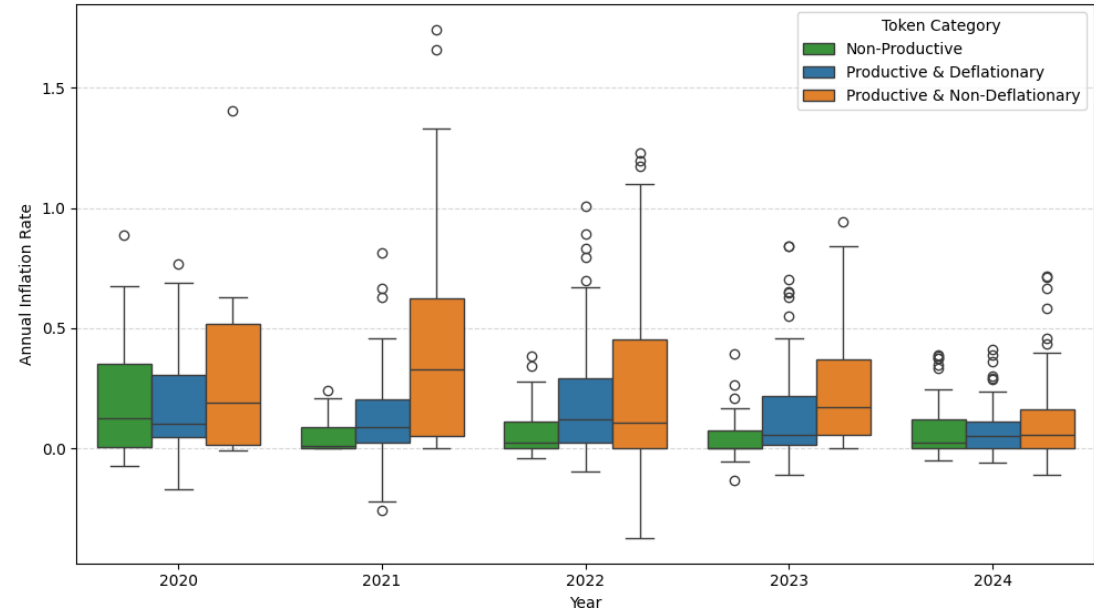


Figure 11. Annual Inflation Rate Distribution by Token Category (2020–2024)

Figure 11 illustrates the annual inflation rate distribution across three token categories—Non-Productive, Productive & Deflationary, and Productive & Non-Deflationary—from 2020 to 2024. The inflation rate measures the rate of expansion in circulating token supply, which, although not identical to the rate of new token issuance, serves as a proxy for short-term changes in the overall supply of tokens within each ecosystem.

Relative to the other two categories, Productive & Deflationary tokens exhibit a much more stable inflation profile over time. During the boom years of 2020–2021, when aggregate crypto prices and activity were elevated, their inflation remains moderate and does not spike as aggressively as that of Productive & Non-Deflationary tokens. In the 2022 downturn, by contrast, their inflation does not collapse: issuance stays positive and, in many cases, is higher than that of Productive & Non-Deflationary tokens, whose inflation is sharply compressed. This pattern is consistent with a more balanced and sustainable incentive structure. Productive & Deflationary systems avoid excessive dilution in exuberant phases, yet continue to provide meaningful rewards to

validators, liquidity providers, and developers when market conditions deteriorate, rather than cutting incentives precisely when participation becomes more fragile. As a result, these ecosystems are able to maintain a continuous, non-extreme flow of token supply and keep the network functioning across market cycles.

Non-Productive tokens, in turn, display consistently low but irregular inflation, with no clear systematic relationship to bull or bear markets. Combined with the absence of corresponding value gains, this indicates that their supply policies are not clearly anchored in ecosystem-expanding activity and function only weakly—if at all—as a coherent incentive mechanism capable of sustaining long-run ecosystem development.

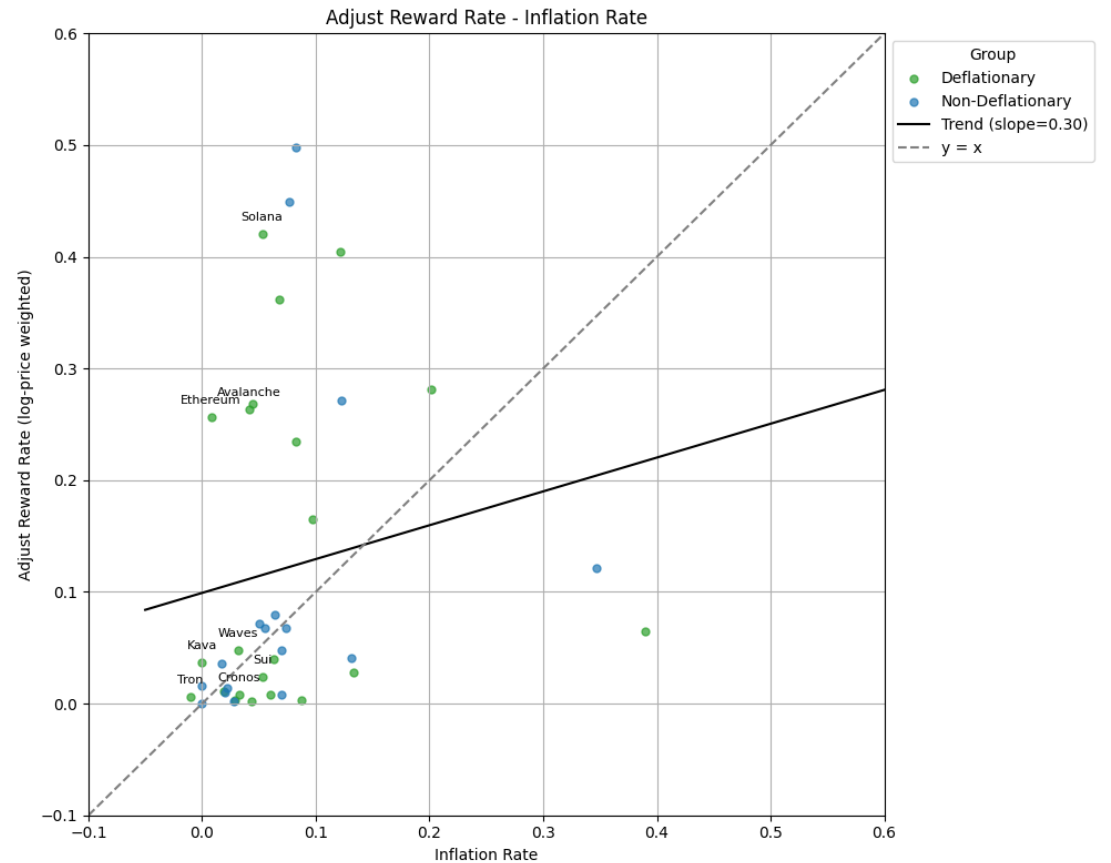


Figure 12. Inflation Rate vs. Staking Reward Rate, 2024.

Note: 1. Inflation rate calculates the annualized expansion rate of an asset’s circulating supply, focusing solely on the current token reward payout and its accrual interval. It offers a clear view of the asset’s short-term supply changes, excluding the total supply from its calculation. Adjust Staking Rewards calculates the current annualized reward rate as a price-weighted average across the network. This metric reflects the average return that stakers can earn—weighted by each asset’s average market price of 2024. 2.Token names labeled in the figure represent the top 10 blockchain networks by Total Value Locked.

Figure 12 presents, for 2024, the relationship between the inflation rate and the adjusted staking reward rate (log-price weighted) for tokens that rely on Proof-of-Stake (PoS) issuance mechanisms. The figure sheds light on the economic incentive structures underlying different blockchain ecosystems. Tokens positioned above the 45-degree

line exhibit adjusted staking rewards that exceed their inflation rates. This indicates that holders are compensated—through additional fee revenues, MEV, or higher token valuations—at a level sufficient to offset dilution from token issuance. Such a configuration is consistent with a relatively active and sustainable ecosystem. By contrast, tokens below the 45-degree line imply that staking returns may be insufficient to counter inflationary pressures, signaling potential long-run value erosion for passive holders. Notably, productive tokens with deflationary mechanisms—such as Ethereum, Solana, and Avalanche—tend to cluster well above the 45-degree benchmark, indicating systematically healthier incentive structures than their non-deflationary peers.

VI. Conclusion

By analyzing 269 blockchain projects (Layer-1s and sidechains) and their token supply mechanisms, this study identifies and classifies two core mechanisms—productive issuance and deflationary design—and proposes the Productive Token Model, which conceptualizes their role as endogenous incentive mechanisms in decentralized systems. The findings show that deflationary mechanisms are primarily embedded in productive tokens, while non-productive tokens with deflationary design are rare and have seen their market share decline year by year. Among different token types, productive-deflationary tokens are particularly prominent in the cryptocurrency market. They exhibit stronger price resilience and liquidity, and their underlying ecosystems show higher capital retention and greater developer concentration. However, they do not necessarily display lower price volatility—in fact, volatility is sometimes higher.

Further analysis indicates that the inflation rates of productive-deflationary tokens remain relatively stable across market cycles. Compared with other token types, their inflation is more moderate during bull markets and does not collapse in bear markets, often remaining at levels that continue to support network participation. By contrast, productive non-deflationary tokens show much greater volatility in inflation rates: inflation tends to surge in bull markets and fall sharply in bear markets. In such systems, the pace of issuance appears less tied to ecosystem demand and more driven by short-term market sentiment, which intensifies reward dilution in booms and leaves producers without institutional support in downturns, thereby increasing the long-term risk of ecosystem contraction. Moreover, the incentive structures of productive-deflationary tokens are more robust: in most cases, staking yields on PoS chains exceed inflation levels. Overall, the core function of the deflationary mechanism is to regulate token supply by reshaping incentives so that continuous issuance follows a smoother and more sustainable path—remaining broadly compatible with ecosystem expansion needs.

In sum, blockchain-based cryptocurrencies constitute a new institutional form of money, with token issuance directly embedded in productive activities and ecosystem expansion, thereby integrating money creation into actual economic processes. Departing from the view that treats cryptocurrencies merely as neutral media of exchange, productive-deflationary tokens actively perform institutional functions in organizing production and mobilizing market expansion. They can be understood as a decentralized evolution of the innovation-mobilization mechanism emphasized in

Schumpeter's Credit Theory of Money—directing seigniorage to ecosystem expanders and maintaining incentive stability through deflationary mechanisms—thus constructing an endogenous growth model that requires neither state sovereignty nor centralized bank-issued credit.

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